VINEETH K

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SUMMARY

Applied AI Engineer who takes automation from problem to production — scoping the processes worth replacing, building the AI and agent systems that do it, integrating them into existing business software, and running them live. Recent work spans retrieval-grounded document assistants, automated invoice workflows, and offline-first field applications with automatic sync.

EXPERIENCE

Applied AI Engineer March 2026 — Present
Loretta Concepts Private Ltd, Thrissur Kerala

- Built an end-to-end RAG-based knowledge assistant for technical documentation using LLMs, OCR, hybrid retrieval, and CrossEncoder reranking. Integrated grounded LLM responses with source/page citations, hallucination controls, and conversation memory.
- Built an intelligent invoice-processing workflow for automated validation, submission, and integration with enterprise systems, reducing manual processing effort.
- Developed an offline-first React Native application for industrial field operations using SQLite, Supabase, PostgreSQL, and automatic synchronization.

AI integrated Data Science Intern June 2025 — November 2025
Techmaghi, Kochi

- Delivered analytics dashboards and visual reports using Power BI and Excel to support business decisions.
- Automated data preprocessing workflows using Python, reducing manual cleaning time.
- Designed ML-based prototypes for prediction and insights generation.
- Collaborated with mentors on real client-oriented datasets to improve analytical accuracy.

EDUCATION

BTECH in ARTIFICIAL INTELLIGENCE & DATA SCIENCE, Jyothi Engineering College 2021 — 2025

PROJECTS

Sign Language Recognition and Translation

- Performed feature extraction, data augmentation, smoothing, and normalization on the raw sensor readings to reduce noise and generate suitable sequential features for machine learning.
- Trained and compared classification algorithms and LSTM networks for gesture classification, with evaluation using a confusion matrix and continuous real-time sensor data.
- Integrated the recognized gestures with a Bluetooth-enabled application for real-time transcription, followed by text-to-speech conversion and multilingual translation using the Google Translate API.

Ball Tracking System

- Developed a YOLOv8-based cricket ball detection and tracking pipeline using Ultralytics YOLO and OpenCV, processing video frame-by-frame and extracting the ball's bounding-box centroid coordinates.
- Implemented trajectory generation and prediction by storing sequential ball coordinates and applying Linear Regression on the latest four points to estimate the ball's next three positions.
- Built an OpenCV-based visualization pipeline that overlays the detected ball path and predicted trajectory onto the final video frame using coordinate points, lines, and markers.

SKILLS

Generative AI : RAG, Hybrid Retrieval, Semantic Search, BM25, CrossEncoder Reranking, OCR, LLM APIs, Voice Agent
AI/ML : Machine Learning, NLP, LSTM, Computer Vision, Object Detection, YOLO, K-Means, Linear Regression
Programming : python, SQL, C
Libraries & Frameworks : Pandas, Numpy, OpenCV, Matplotlib, Seaborn, FastAPI, SQLAlchemy
Databases : PostgreSQL, redis, Supabase
Data Analytics : Data Cleaning, Exploratory Data Analysis, Power BI
Tools & Platforms : Git, Github, Jupyter Notebook, Visual Studio Code, Google Colab, Vapi

PUBLICATION

Enhancing Real-Time Communication with a Smart Glove: LSTM Based Sign Language Recognition
2025 4th International Conference on Advances in Computing, Communication, Embedded and Secure Systems (ACCESS)
DOI : https://doi.org/10.1109/ACCESS65134.2025.11135890
Presented an LSTM-based smart glove system for real-time sign language recognition and translation, improving accessibility and enabling faster human–machine communication.